Distributed linear quadratic tracking control for leader-follower multi-agent systems: a suboptimality approach

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Applications

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Figure: Platooning of intelligent transportation systems

Figure: Formation flying of planes

Motivation

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- the influence of external disturbances to the system behaviors (H_2 or H_∞ sense),
- the energy consumption (LQ sense).

Motivated by the above, in this talk, we consider the problem of distributed LQ tracking control for leader-follower multi-agent systems.

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Distributed LQ tracking control problem

We consider a leader-follower multi-agent system consists of

- **One leader:**

$$\dot{x}_r(t) = Ax_r(t), \quad x_r(0) = x_{r0} \tag{1}$$

- N **followers** ($N \geq 1$)**:**

$$\dot{x}_i(t) = Ax_i(t) + Bu_i(t), \quad x_i(0) = x_{i0}, \quad i = 1, 2, \ldots, N \tag{2}$$

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Assumptions:

- (A, B) is stabilizable.
- At least one follower receives the state information of the leader.
- The communication graph between the followers is a connected simple undirected graph.

Distributed LQ tracking control problem

The agents are to be interconnected by the distributed diffusive control law

$$u_i(t) = K \sum_{j=1}^{N} a_{ij}(x_i(t) - x_j(t)) + K g_i(x_i(t) - x_r(t)), \tag{3}$$

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- a_{ij} is the ij-th entry of the adjacency matrix of the communication graph between the followers.
- $g_i > 0$ if the leader shares its state information with the i-th agent.

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Definition 1

The control law (3) achieves tracking consensus for the leader-follower system (1) and (2) if for all $i = 1, 2, \ldots, N$ and for all initial states x_{r0} and x_{i0}, we have

$$x_i(t) - x_r(t) \to 0 \text{ as } t \to \infty.$$

Distributed LQ tracking control problem

Furthermore, we introduce the global LQ cost functional

$$J(u) = \int_0^\infty \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} a_{ij}(x_i - x_j)^\top Q(x_i - x_j) + \sum_{i=1}^{N} g_i(x_i - x_r)^\top Q(x_i - x_r) + \sum_{i=1}^{N} u_i^\top R u_i \; dt, \qquad Q \geq 0, \; R > 0 \tag{4}$$

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Distributed LQ optimal tracking control problem

Find gain matrix K such that the cost functional (4) is minimized for all initial states x_{r0} and x_{i0}, $i = 1, 2, \ldots, N$ over all distributed diffusive control laws that achieve tracking consensus.

Distributed LQ tracking control problem

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Therefore, in this paper we will solve a suboptimality version of this problem.

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It is difficult to find a closed-form solution, or such optimal controller may not even exist.

Therefore, in this paper we will solve a suboptimality version of this problem.

Next, we will formulated the distributed suboptimal LQ tracking problem.

Distributed LQ tracking control problem

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Subsequently, the dynamics of the error state is given by

$$\dot{e}_i = Ae_i + Bu_i, \quad i = 1, 2, \ldots, N. \tag{5}$$

Distributed LQ tracking control problem

Denote

$$x = \left(x_1^\top, \ldots, x_N^\top\right)^\top,$$

$$u = \left(u_1^\top, \ldots, u_N^\top\right)^\top,$$

$$e = \left(e_1^\top, \ldots, e_N^\top\right)^\top.$$

Distributed LQ tracking control problem

Error system:

$$\dot{e} = (I_N \otimes A)e + (I_N \otimes B)u, \quad e(0) = e_0. \tag{6}$$

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Distributed control law:

$$u(t) = (\Gamma \otimes K)e, \tag{7}$$

where

$$\Gamma = L + G, \quad G = \mathrm{diag}(g_1, g_2, \ldots, g_N).$$

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Cost functional:

$$J(u) = \int_0^\infty e^\top(\Gamma \otimes Q)e + u^\top(I_N \otimes R)u \; dt. \tag{8}$$

Distributed LQ tracking control problem

Closed-loop error system:

$$\dot{e} = (I_N \otimes A + \Gamma \otimes BK)e, \quad e(0) = e_0 \tag{9}$$

Cost:

$$J(K) = \int_0^\infty e^\top \Big(\Gamma \otimes Q + \Gamma^2 \otimes K^\top R K\Big) e \; dt \tag{10}$$

The controlled leader-follower system reaches tracking consensus if and only if the closed-loop error dynamics (9) is stable.

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Distributed LQ suboptimal tracking problem

Let $r > 0$ and $\gamma > 0$. Find gain matrix K such that the controlled leader-follower system reaches tracking consensus and the associated LQ cost satisfies $J(K) < \gamma$, for all initial conditions x_0 and x_{r0} such that $e_0 = x_0 - x_{r0}$ satisfies

$$B(r) = \{e_0 \in \mathbb{R}^{nN} \mid \|e_0\| \leq r\}.$$

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Brief ideas

1. We apply a state transformation

$$\bar{e} = (U^\top \otimes I_n)e$$

with $U^\top \Gamma U := \Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_N), \Gamma = L + G$. We transform the problem of suboptimal tracking consensus into the problems of LQ suboptimal control for N auxiliary systems (in which λ_i appears) with the same control gain K.

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2. As a key step, we provide a necessary and sufficient condition for a given control gain K to stabilize all auxiliary systems and to guarantee $J < \gamma$.

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2. As a key step, we provide a necessary and sufficient condition for a given control gain K to stabilize all auxiliary systems and to guarantee $J < \gamma$.

3. Based on the result in Step 2, as the main result, we provide a design method for computing control gain K such that the resulting distributed control law achieves tracking consensus and guarantees $J < \gamma$.

Main result

Theorem 1

Let $r > 0$ and $\gamma > 0$. Let λ_1 and λ_N be the smallest and largest eigenvalue of $\Gamma = L + G$. Let c be any real number such that $0 < c < \frac{2}{\lambda_N}$. We distinguish two cases:

1. if $\frac{2}{\lambda_1+\lambda_N} \leq c < \frac{2}{\lambda_N}$, then there exists $P > 0$ satisfying

$$A^\top P + PA + (c^2\lambda_N^2 - 2c\lambda_N)PBR^{-1}B^\top P + \lambda_N Q < 0. \tag{11}$$

2. if $0 < c < \frac{2}{\lambda_1+\lambda_N}$, then there exists $P > 0$ satisfying

$$A^\top P + PA + (c^2\lambda_1^2 - 2c\lambda_2)PBR^{-1}B^\top P + \lambda_N Q < 0. \tag{12}$$

Main result

Theorem 1 (continued)

In both cases, if in addition P satisfies

$$P < \frac{\gamma}{r^2} I, \tag{13}$$

then the distributed control law (3) with $K := -cR^{-1}B^\top P$ achieves tracking consensus and $J(K) < \gamma$ for all initial states x_{r0} and x_0 satisfying

$$x_0 - \mathbf{1}_N \otimes x_{r0} \in B(r). \tag{14}$$

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In summary,

- We have formulated the distribute LQ suboptimal tracking control problem for leader-follower multi-agent systems.
- We have provided a design method for computing distributed suboptimal control laws that achieve tracking consensus and guarantee the associated cost is smaller than a given tolerance for all initial states bounded in norm by a given radius.

Thank you very much for your attention!